

Operational Quality Management

Science

May, 2026

Operational Quality Management (A33461)

Short Title:	Operational QM	
Faculty	Science and Computing	
Department	Science	
Cluster	Quality, Regulatory and Compliance	
Author	BWHELAN	
Credits	10	Level: Advanced

Description of Module / Aims

The aim of this module is to provide the professional adult learner with knowledge relating to quality and operations management, lean manufacturing, six sigma, leadership, strategy and competitive advantage which they can apply to their workplace environment.

Programmes

QUAL - 0034	BSc (Hons) in Good Manufacturing Practice and Technology (WD_SGMPT_B)	4 / 2 / M
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Indicative Content

- Lean Management Systems, Six Sigma, Operational Excellence
- Strategy and Leadership
- Organisational culture and management
- Change Management, global context for change
- Performance Management

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Appraise the application of Lean Six Sigma and associated systems in the workplace.
2. Critique the impact of strategic management and effective leadership.
3. Evaluate adaptive organisational culture.
4. Assess change management initiatives using case studies.
5. Interpret the application of performance metrics to maximise competitive advantage.

Learning and Teaching Methods

- Lectures
- Case studies
- Group discussion of articles
- Seminars
- Independent study

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	1, 2, 3, 4, 5
<i>Assignment</i>	50%	1, 2, 3
<i>Case Studies</i>	30%	4
<i>Presentation</i>	20%	5

Assessment Criteria

- < 40%:** Little or no demonstration of an understanding of the fundamentals of the learning outcomes. Lack of competence with associated laboratory skills & practices & an inability to carry out critical thinking.
- 40%--49%:** Will show a limited understanding of the fundamentals of the learning outcomes. May have some difficulty with applying critical thinking and problem-solving. Will display some ability with associated laboratory skills & practices.
- 50%--59%:** As well as the above, the candidate will show a good understanding of the fundamentals of the learning outcomes with demonstration of some critical thinking & problem-solving skills. Will display good competence with associated laboratory skills & practices.
- 60%--69%:** In addition, will demonstrate more detailed knowledge of all learning outcomes, very good critical thinking skills & a high level of competence & efficiency in associated laboratory practicals.
- 70%--100%:** All the above to an excellent level. In addition, will demonstrate a comprehensive knowledge & understanding of all learning outcomes & be able to discuss the area. Will display advanced competence & independence with associated laboratory skills & practices.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture		24
Independent Learning		246

Essential Material

- Burnes, B. *Managing Change: a strategic approach to organisational dynamics*. 7th. XX: Pearson, 2017.
- Mann, D. *Creating a Lean Culture: Tools to sustain Lean Conversions*. 3rd. XX: Productivity Press, 2015.

Supplementary Material

- "Harvard Business Review." <https://hbr.org>
- Jeffs, C. *Strategic Management*. 1st. XX: Sage Publications, 2012.
- Schein, E. *Organisational Culture & Leadership*. 5th. XX: Wiley & Sons, 2017.